

Section C — Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

Objective:

At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

Ans.:-

Task 1 :- Identify the three hardware components most likely to cause random reboots under load.

Ans.:-

CPU (Processor) – Can overheat during heavy workloads and cause automatic shutdown or restart.

GPU (Graphics Card) – Rendering creates high GPU load and heat, which can cause instability or rebooting.

PSU (Power Supply Unit) – An insufficient or failing PSU may not provide stable power when CPU and GPU demand increases.

Task 2:- Explain why the systems pass POST in the morning but fail in the afternoon.

Ans.:-

Morning: The workstations are cool and under low workload, so CPU, GPU, and PSU operate normally and successfully pass POST.

Afternoon: Heavy rendering increases CPU and GPU temperature and power consumption. The systems are kept inside poorly ventilated cabinets, so hot air cannot escape. As temperature rises, the CPU/GPU may reach unsafe levels and trigger thermal protection, causing random restarts. Increased power demand can also expose a weak or failing PSU, causing unstable power and reboots.

Task 3 :- Describe how you would use a stress-test tool to confirm a thermal or power issue.

Ans.:-

1. Use a tool such as OCCT, AIDA64, or Prime95 to stress the CPU/GPU.
2. Monitor CPU/GPU temperature, fan speed, and PSU voltage while the test runs.
3. Run the test for 15–30 minutes under heavy load.
4. If temperature rises excessively and the PC reboots, it indicates a thermal problem.
5. If temperatures remain normal but the PC suddenly restarts during high CPU + GPU load, suspect a PSU/power-delivery problem.
6. Repeat the test after improving cooling or using a known-good PSU to confirm the cause.

Task 4 :- Propose three specific steps to remediate the frequent system restarts.

Ans.:-

1. Improve ventilation:-

Cut intake and exhaust ports directly into the desk cabinets. Install AC-powered or USB-powered active ventilation fans to break up the trapped hot micro-climate and continuously cycle fresh room air through the cabinet.

2. Clean and service cooling:-

Open the workstation chassis to completely clear out all dust blockages from the heat sinks, intake filters, and cooling fans using compressed air. Remove the existing, degraded thermal interface material from the CPU and GPU, and re-apply a high-performance carbon-based thermal paste to maximize heat transfer to the coolers.

3. Check the PSU:-

Boot each workstation into the BIOS/UEFI menu and modify the PWM system fan settings to a highly aggressive fan curve. Configure the CPU, GPU, and chassis fans to ramp up to 100% speed much earlier rather than waiting for the components to near their critical thermal limits.

Task 5 :- Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Ans.:-

A PSU upgrade is necessary if the workstations suffer from thermal derating due to the trapped cabinet heat. A PSU upgrade may be necessary if the existing PSU is underpowered, faulty, old, or unable to provide stable voltage during heavy CPU/GPU workloads. Rendering increases power consumption, which can cause sudden restarts if the PSU cannot handle the load.

A PSU upgrade may not be necessary if the current PSU has sufficient wattage, is in good condition, and remains stable during stress testing. In that case, the main cause is more likely overheating due to poor ventilation.